

COMMUNICATION LABORATORY

Experiment 3: Binary Phase Shift Keying (BPSK) Modulation and Demodulation Scheme



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TITLE: Design of Binary Phase Shift Keying (BPSK) Modulation and Demodulation scheme

1. **PROBLEM STATEMENT:** Using the PN sequence generated in the previous experiment as data:

1. Design and implement a circuit for the working of modulator and demodulator for the **binary phase shift keying (BPSK)**.
 2. Extract the BPSK modulated data stream, and using MATLAB, demodulate the original data stream (in software domain). Now perturb the BPSK modulated data by adding appropriate noise (additive Gaussian noise) to simulate transmission over an additive Gaussian noise channel, and attempt to demodulate with the same demodulator (take three different noise variance values). Comment on your observation.
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2. EQUIPMENT LIST & KEY COMPONENTS:

1. **IC 741 (Op-Amp):** Used for the inverting amplifier, adder, and voltage comparator circuits.
2. **IC 7404 (NOT Gate):** Used to invert the digital data signal.
3. **IC M4016 (Quad Bilateral Switch):** Used as the switching modulator.
4. **IC 7495 (4-bit shift register) / IC 74164 (8-bit shift register):** Used for PN sequence generation.
5. **IC 7486 (Quad 2-input XOR gate):** Used in the PN sequence generator feedback loop.
6. **Passive Components:** Resistors (1k Ω etc.), Capacitors (1 μ F), Breadboard, Connecting wires.
7. **Silicon Diode (1N4148 or equivalent):** Used in the envelope detector circuit for rectification (with a typical 0.7V forward bias drop).
8. **Instruments:** Digital Storage Oscilloscope (DSO), Signal Generator, Power supply (+5 / -5 / +12 / -12 / GND), Multimeter.

(See Appendix A for IC PIN Diagrams).

3. BRIEF THEORY & BACKGROUND

3.1 Why Modulation?

Modulation is the process of altering a reference signal with a fixed frequency (the carrier wave) to transmit data. In digital modulation, basis functions are chosen for a particular modulation scheme, generally orthogonal to each other (derived using the Gram-Schmidt orthogonalization procedure) [1]. Once basis functions are chosen, any vector in the signal space can be represented as a linear combination of them. Modulation allows for frequency multiplexing, reduces necessary antenna sizes, and provides robustness against noise and interference during transmission [3].

3.2 What is Binary Phase-Shift Keying (BPSK)?

Phase-shift keying (PSK) is a digital modulation technique that transmits data by altering the phase of the carrier wave. The sine and cosine inputs are changed at precisely the right time to produce the modulation [2]. It is frequently used for Bluetooth, RFID, and wireless LAN communication.

The most basic type of PSK is BPSK (sometimes referred to as PRK, phase reversal keying, or 2-PSK) because it employs two phases that are 180° apart. In BPSK, there is only one basis function, which is a sinusoid: $\phi_1(t) = (\sqrt{2}/T_b) \cos(2\pi f_c t)$ [1][2].

We use the following signals to denote bit 0 and bit 1:

- $s_1(t) = A_c \cos(2\pi f_c t)$, $0 \leq t \leq T_b$ for binary 1
- $s_2(t) = A_c \cos(2\pi f_c t + \pi) = -A_c \cos(2\pi f_c t)$, $0 \leq t \leq T_b$ for binary 0

Where A_c is the amplitude of the sinusoidal signal, f_c is the carrier frequency (in Hz), t is the instantaneous time, and T_b is the bit duration [1].

We can intuitively recognize that the system will be more robust if there is greater separation between these two phases. The maximum difference between the logic-high and logic-low phases is 180°. Shifting a sinusoid by 180° is the same as inverting it; thus, we can think of BPSK as simply inverting the carrier in response to one logic state and leaving it alone in response to the other [3].

The constellation points are depicted at 0° and 180° on the actual axis (at $+\sqrt{E_b}$ and $-\sqrt{E_b}$). As a result, BPSK handles the highest amount of noise or distortion before the demodulator makes a mistake. The theoretical probability of bit error over an AWGN channel is low, given by the Q-function [2]: $P_e = Q(\sqrt{2E_b}/N_0)$

3.3 PN Sequence as Input Data

The data used throughout this experiment is the PN sequence generated in Experiment 1. A Pseudo-Noise (PN) sequence is a deterministic binary sequence generated by a Linear Feedback Shift Register (LFSR). Although deterministic, a

maximal-length PN sequence has statistical properties similar to a truly random binary sequence [1]. For an LFSR of degree $m=5$, the maximum period is $N=2^5-1=31$, provided the feedback polynomial is primitive (e.g., $P(x)=x^5+x^2+1$) [2]. This 31-bit sequence provides a rich, balanced stream of 1s and 0s, ideal for evaluating the transition performance of the BPSK modulator.

3.4 Advantages and Disadvantages of BPSK

Advantages:

- It is the most reliable modulation technology since binary 1 and 0 are separated by a 180-degree carrier phase shift. This allows BPSK modulated data to be sent from base stations over greater distances. [3]
- Pilot carrier and preamble sequences use BPSK modulation for channel estimation, equalization, and time/frequency synchronization.
- The receiver is quite straightforward, making it highly power-efficient. [1]

Disadvantages:

- One bit is carried by a single analog carrier. As a result, the symbol rate and data rate are the same (1 bit per symbol). [2]
- It is far less bandwidth-efficient than higher-order modulation strategies like QPSK, 16QAM, and 64QAM. It cannot be used for large data-rate applications. [1][2]

4. LINE CODING TECHNIQUES

4.1 Purpose of Line Coding

Line coding converts digital data into baseband signals suitable for transmission. Its primary goals include ensuring signal integrity against channel distortions, embedding clock synchronization data to prevent misinterpretation, eliminating DC components (baseline wander), and optimizing bandwidth spectrum utilization. [1][3]

4.2 Classifications and Schemes

Line codes are generally categorized by their voltage levels and transition rules [1][3]:

- **Unipolar & Polar:** Unipolar uses a single non-zero voltage level (e.g., Unipolar NRZ), whereas Polar uses both positive and negative voltages, which helps eliminate DC bias (e.g., Polar NRZ, Manchester).
- **Bipolar & Multilevel:** Schemes like AMI use three levels to embed error-checking, while MLT-3 improves bandwidth efficiency by reducing the fundamental transition frequency.

- **NRZ vs. RZ:** Non-Return-to-Zero (NRZ) schemes maintain the voltage level for the full bit period to maximize bandwidth efficiency. Conversely, Return-to-Zero (RZ) schemes drop to zero mid-bit, which costs bandwidth but greatly aids the receiver in clock recovery.

4.3 Line Coding in BPSK (Polar NRZ)

In this experiment's hardware implementation, standard binary unipolar logic (0V and +5V) is used to control the bilateral switches. However, fundamentally, BPSK is mathematically equivalent to modulating the carrier with a **Polar Non-Return-to-Zero (NRZ)** line code (+1 and -1) [2]. The switching mechanism in the modulator circuit effectively translates the unipolar control logic into this polar phase shift by selecting between a 0° and 180° carrier for the entire bit duration (NRZ). [3]

Typical Plots of Explored Line Codes:

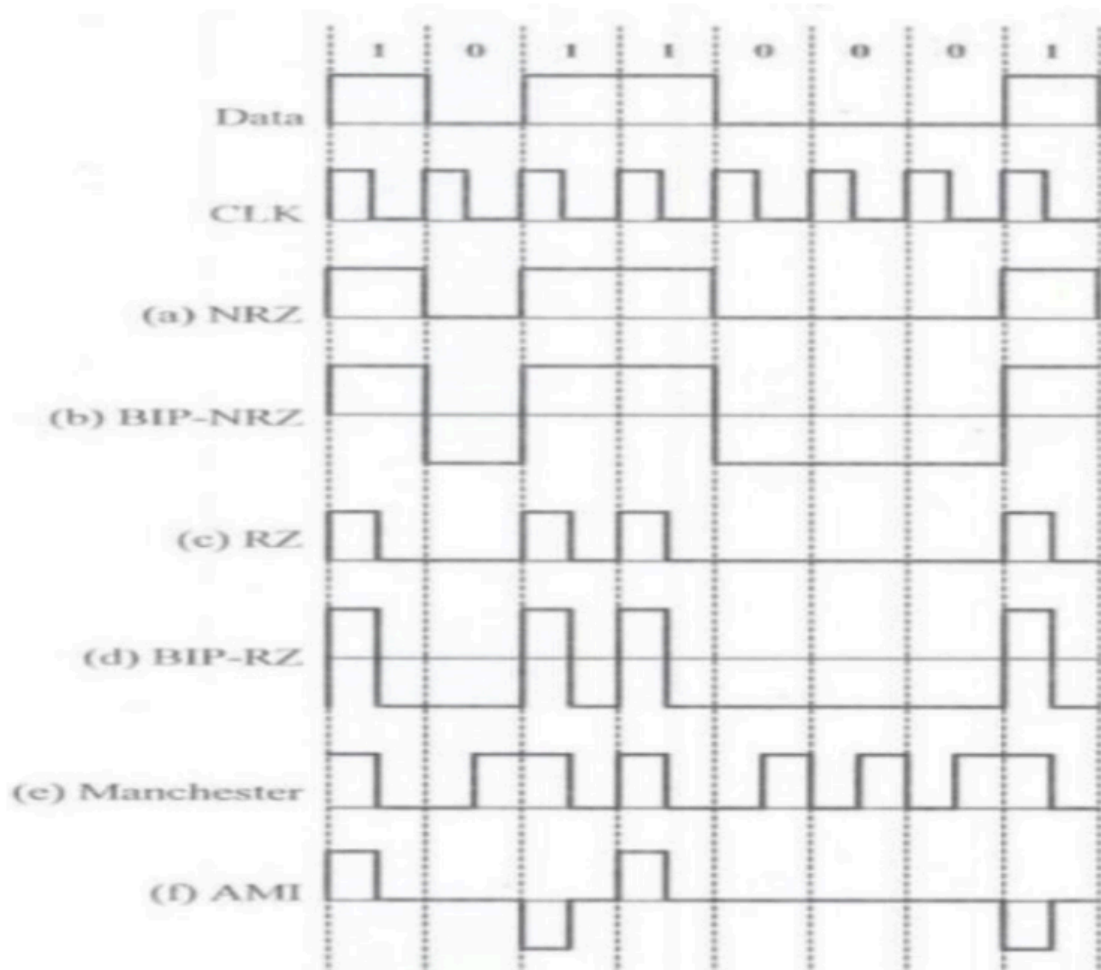


Fig 4.1 : Waveforms of various line coding techniques (NRZ, RZ, Manchester, AMI)

5. DESIGN & CIRCUIT IMPLEMENTATION

5.1 Block Diagrams & Circuit Diagrams

(Insert Modulator Block Diagram here) (Insert Modulation Circuit Schematic here)
(Insert Demodulation Circuit Schematic here)

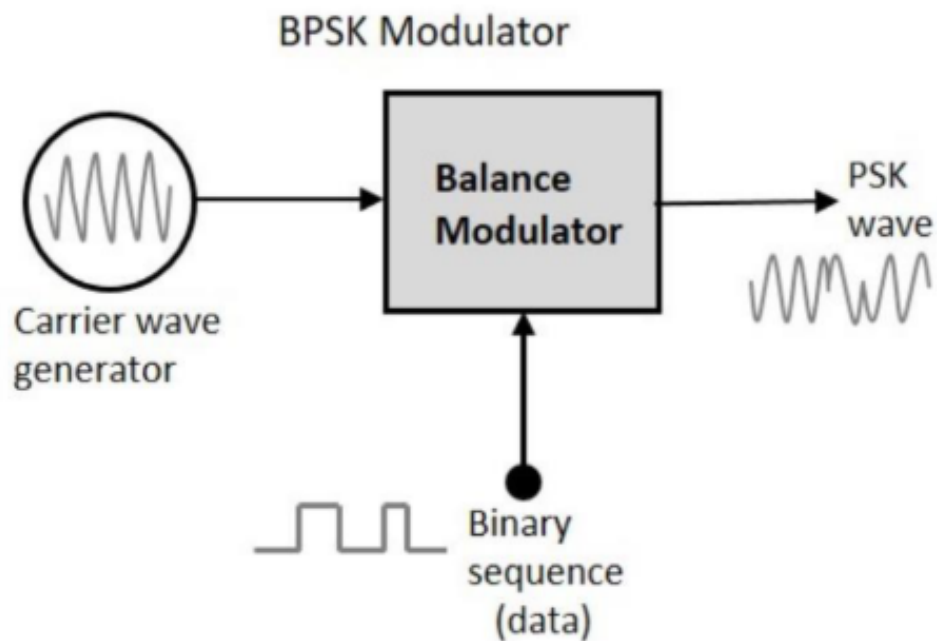


Fig 5.1.1 : Modulator Block Diagram

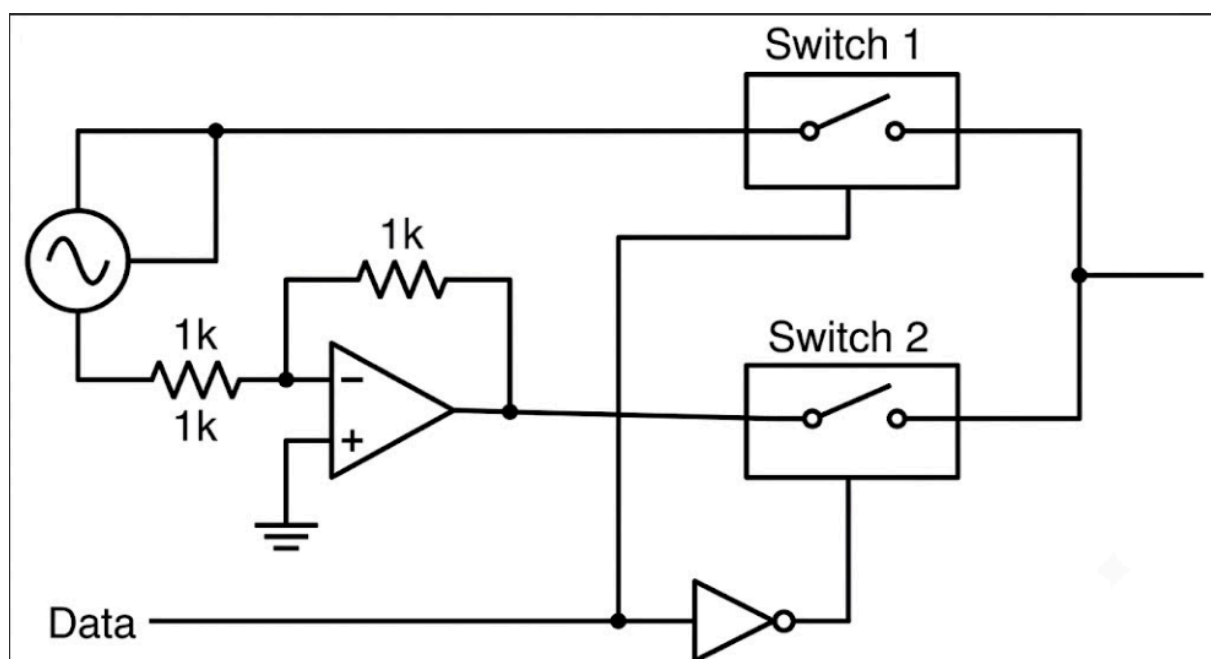


Fig 5.1.2 : Modulation Circuit Diagram where the Data is the PN sequence

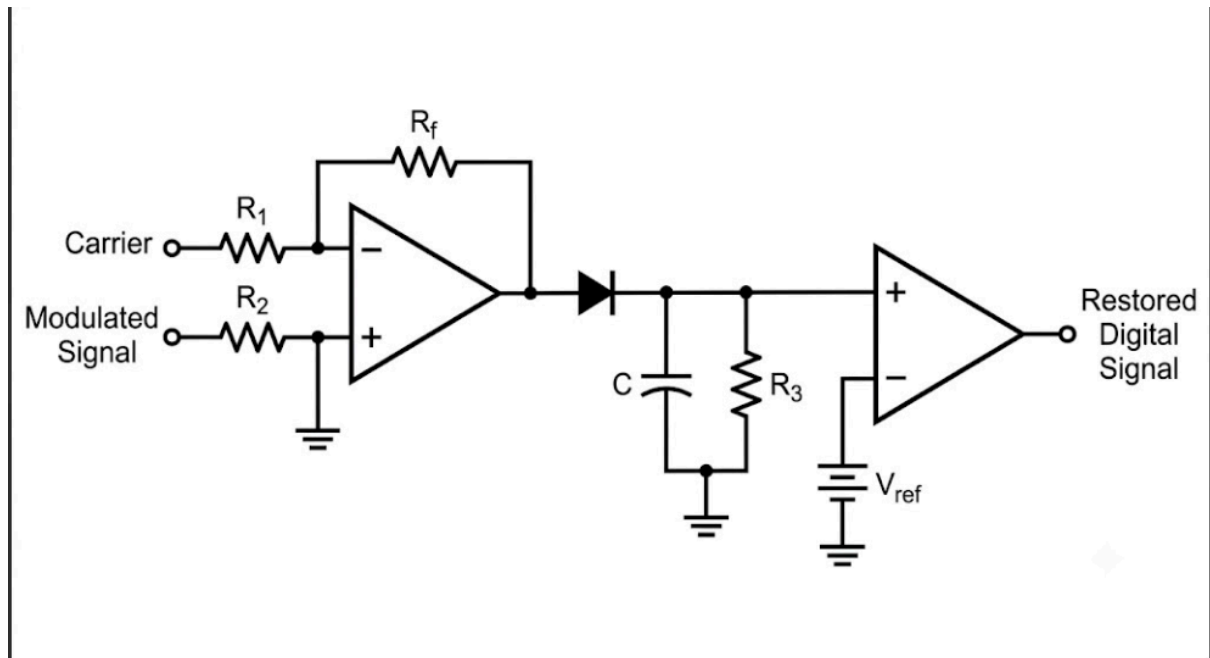


Fig 5.1.3 : Demodulation Circuit Diagram where all the resistors (R_1 , R_2 , R_3 , R_f) are $1k$ and the capacitor C is $1\mu F$ and the V_{ref} is $100mV$ (took it from voltage divider circuit)

5.2 Modulator Circuit Operation (Using CD4016)

The modulation of BPSK is done using a balance modulator. For this hardware lab, we achieve this using the **IC M4016 Quad Bilateral Switch**.

Signal Generation and Routing: Let $C_1 = \sin(\omega_c t)$ and $C_2 = -\sin(\omega_c t)$.

1. A base carrier $\sin(\omega_c t)$ at 4 kHz with $1V_{pp}$ is generated from the function generator.
2. The 180° shifted carrier $-\sin(\omega_c t)$ is actively generated by passing the base carrier through an **Inverting Op-Amp** (IC 741). By using $1\text{ k}\Omega$ resistors for both the input (R_{in}) and feedback (R_f) paths, the amplifier achieves a precise unity gain inversion ($\text{Gain} = -R_f/R_{in} = -1$).
3. The PN sequence is driven by a digital clock set to 1 kHz at $5V_{pp}$ ($0V$ to $+5V$ TTL logic). This creates a perfect harmonic relationship: with a 1 kbps data rate and a 4 kHz carrier, there are exactly 4 complete carrier cycles per bit period, ensuring clean phase transitions at the bit boundaries.
4. The original data sequence (PN) is fed to the control pin of Switch 1 (which gates $\sin(\omega_c t)$).
5. The inverted data sequence (NOT PN, via IC 7404) is fed to the control pin of Switch 2 (which gates $-\sin(\omega_c t)$).

Output Node Logic (Breadboard Shorting): Because the PN and NOT PN sequences are logically mutually exclusive, one switch is strictly ON while the other is strictly OFF. This allows us to simply tie both switch outputs to the *exact same line on the breadboard*. Because the same line has the same potential, and there is no

contention between the switches, the combined output inherently produces the final alternating-phase BPSK signal without the need for a secondary summing amplifier.

5.3 Demodulator Circuit Operation

The hardware demodulation relies on a non-coherent envelope detection technique [3].

1. **The Adder (Op-Amp):** We added the original carrier signal C_1 and the modulated signal $VBPSK$ using an IC 741 Op-Amp adder circuit [1].
 - When $D=1$, $V_{BPSK}=C_1$. Adder Output: $V_{add}=C_1+C_1=2C_1$.
 - When $D=0$, $V_{BPSK}=-C_1$. Adder Output: $V_{add}=-C_1+C_1=0$.
 - This operation seamlessly converts the constant-envelope BPSK signal into an Amplitude Shift Keying (ASK) / On-Off Keying varying envelope, where one state has zero amplitude and the other has double amplitude [3].
2. **Envelope Detector:** The ASK output is passed through a diode-RC circuit. The standard Silicon Diode (1N4148) rectifies the signal, introducing a standard $\approx 0.7V$ forward bias voltage drop, which significantly attenuates the absolute peak voltage entering the RC stage. The RC parallel combination then traces the peak envelope. We specifically tuned the circuit using $R=1k\Omega$ and $C=1\mu F$. It charges rapidly during the high-amplitude carrier bursts and discharges exponentially during the low periods.
3. **Voltage Comparator:** After the envelope detector, the resulting peak output was approximately 114 mV. This bumpy envelope is passed into an IC 741 configured as a comparator. Specifically, the envelope detector's output was fed into the **non-inverting input (+)** of the Op-Amp, while a constant DC reference voltage (V_{ref}) of 100 mV was applied to the **inverting input (-)**. When the envelope voltage exceeds 100 mV, the comparator swings to positive saturation (logic HIGH); when it falls below, it swings to negative saturation (logic LOW). This thresholds the signal and restores the crisp digital square wave.

5.4 Expected Outcome

1. **Modulator Expected Outcome:** We expect to see a continuous, constant-amplitude sinusoidal carrier whose phase shifts abruptly by 180° exactly at the transitions of the input digital square wave.

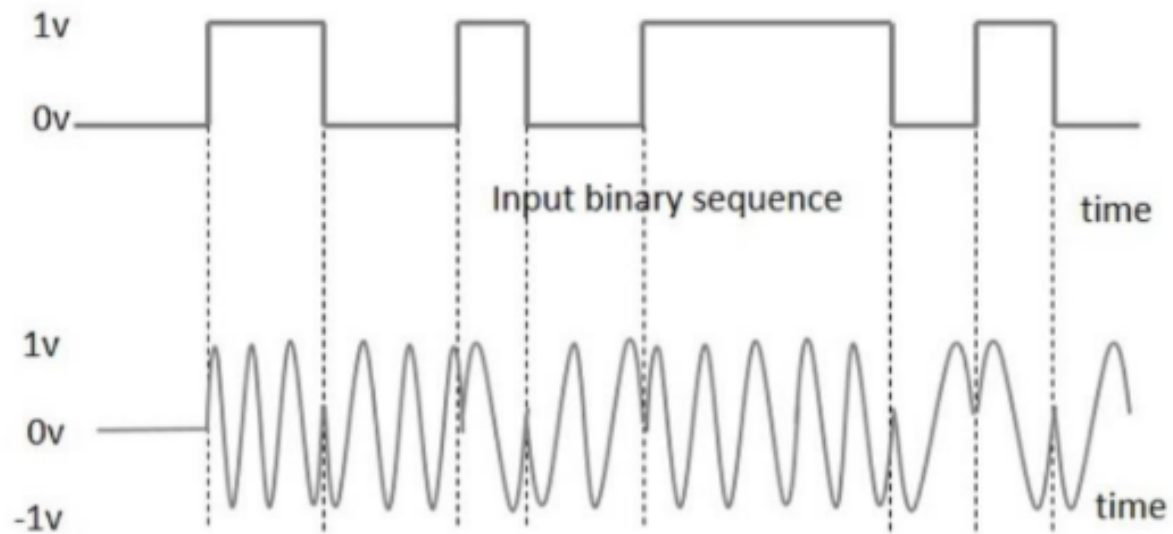


Fig 5.4.1 : The expected Modulated Output

2. Demodulator Expected Outcome:

- Post-adder: The signal should appear as an ASK wave (high amplitude bursts separated by near-zero amplitude).
- Post-envelope detector: The wave should display classical RC charging (sharp rise) and discharging (exponential decay) curves tracking the high-amplitude carrier bursts.
- Final Output: A restored square wave perfectly matching the original PN sequence but slightly delayed in time due to the RC filter.

6. EXPERIMENT RESULTS & PLOTS

6.1 Hardware Results



Fig 6.1.1 : Modulated BPSK Output - 1



Fig 6.1.2 : Modulated BPSK signal - 2

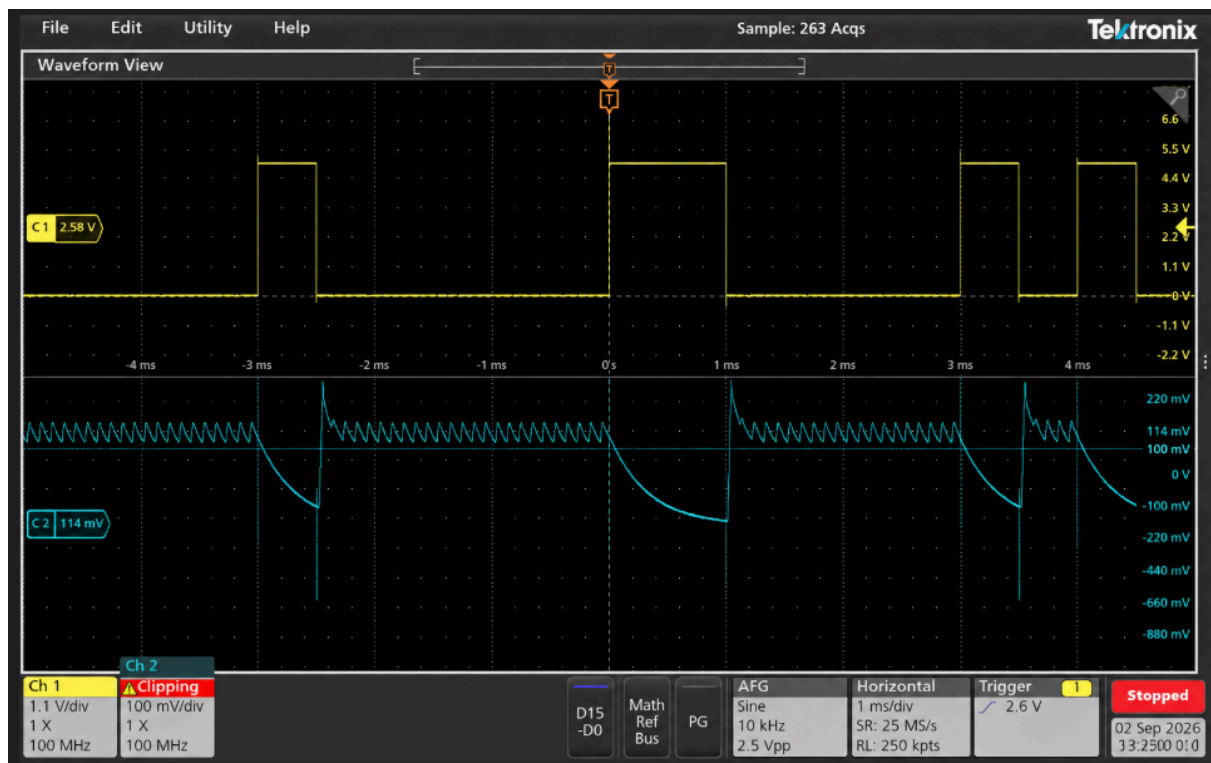


Fig 6.1.3 : Output at the Node of Envelope Detector



Fig 6.1.4 : Demodulated Output - 1



Fig 6.1.5 : Demodulation Output - 2

6.2 Software Results (MATLAB Simulation)

Using the extracted BPSK data from the DSO, a software-defined receiver was implemented in MATLAB to decode the signal, inject Additive White Gaussian Noise (AWGN), and calculate the Bit Error Rate (BER).

BPSK Modulation and Software Recovery



Fig 6.2.1 : Software Recovered Data vs Original

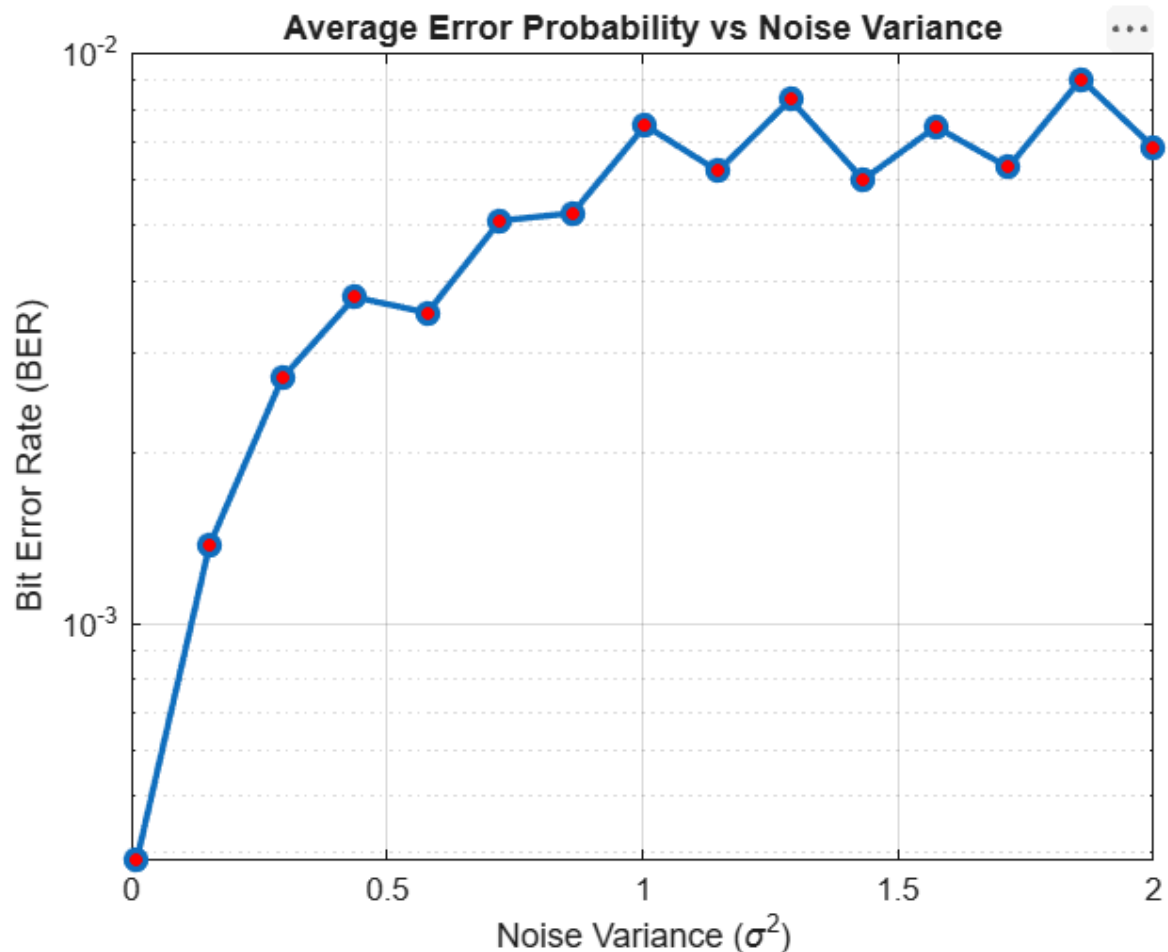


Fig 6.2.2 : Average Error Probability vs Noise Variance curve

7. INFERENCES & DISCUSSION

1. **Analog vs. Digital Inversion Requirements:** A critical distinction in the circuit design was how signal inversion was handled for different domains. The PN sequence is a digital logic signal (0V/+5V), so it could be easily and instantly inverted using a standard digital TTL NOT gate (IC 7404). However, the 4 kHz carrier is a continuous analog sine wave. A digital NOT gate cannot invert an analog signal (it would simply clip it into a distorted square wave). Thus, to generate the precise 180° phase-shifted carrier ($-\sin x$), we had to construct an active analog inverting amplifier using an IC 741 Op-Amp with a gain of -1.
2. **Breadboard Multiplexing Logic:** During modulation, because the PN and NOT PN control signals perfectly complement each other, there was never a state where both switches were active simultaneously. This allowed us to elegantly multiplex the output by directly shorting the two switch outputs on the same breadboard row, eliminating the need for an extra summing amplifier stage and avoiding voltage collision.
3. **Carrier Amplitude Limitation ($1V_{pp}$):** The base carrier signal amplitude was

deliberately restricted to $1V_{pp}$ (swinging from +0.5 V to -0.5 V). This limitation is driven by the IC M4016 bilateral switch; passing analog signals with large negative voltage swings can forward-bias the IC's internal protective substrate diodes (especially if operated on standard unipolar logic supplies), leading to severe signal clipping and distortion.

4. **Carrier Frequency Tuning (1kHz vs 4kHz):** Initially, the experiment was attempted using a 1 kHz carrier signal at $1V_{pp}$. However, at this frequency, the output of the envelope detector was not distinguishable enough to act as a reliable input for the voltage comparator. To resolve this, we increased the carrier frequency to 4 kHz. A higher carrier frequency allows for more cycles per bit period, yielding a much tighter and more distinguishable envelope for the comparator to trigger on.
5. **RC Time Constant Optimization:** For the envelope detector to properly trace the signal, the fundamental condition $f_c > 1/2\pi RC$ must be met. Using $R=1k\Omega$ and $C=1\mu F$, we achieved a time constant of $\tau=1$ ms, or a cutoff frequency of approximately 159 Hz. Since our adjusted carrier $f_c=4000$ Hz is much greater than 159 Hz, the discharging profile was perfect (**as visually confirmed in Fig 6.1.3**), avoiding both excessive ripple (too fast discharge) and diagonal clipping (too slow discharge).
6. **Comparator Threshold and Duty Cycle Distortion:** The peak output voltage directly after the envelope detector was measured at around 114 mV. We consequently set the comparator reference voltage (V_{ref}) to 100 mV. Because this V_{ref} threshold was set quite high relative to the 114 mV peak, the restored square wave (**Fig 6.1.4 and Fig 6.1.5**) did not have a perfectly 50% duty cycle (a larger portion of the transition slope fell below the threshold). If the reference voltage had been lowered perfectly to the midpoint of the envelope swing, the output square wave would have been a perfectly proportioned reconstruction of the original PN sequence.
7. **Software BER Simulation:** As plotted in Fig 6.2.2, the MATLAB simulation successfully demonstrates the effect of an AWGN channel on BPSK. At low noise variances, the simulated envelope detector easily distinguishes the signal from the noise (yielding a near-zero BER). However, as the noise variance increases past 1.0, the random Gaussian noise peaks frequently push the signal across the decision threshold during low-amplitude states, causing the Bit Error Rate (BER) to steadily climb, perfectly reflecting theoretical AWGN channel degradation.

8. CONCLUSION

The BPSK modulation and demodulation scheme was successfully designed and implemented using discrete logic gates, analog bilateral switches (IC M4016), and operational amplifiers (IC 741). The experiment verified that switching between two out-of-phase carriers effectively produces a robust BPSK signal. Non-coherent demodulation via an Op-Amp adder and envelope detector was proven to be a viable, cost-effective receiver architecture. Furthermore, experimental tuning—such as adjusting the carrier to 4 kHz and optimizing the comparator threshold relative to the

114 mV envelope peak—was critical for accurate signal recovery. Finally, the MATLAB simulation successfully reinforced the theoretical robustness of BPSK, illustrating the direct relationship between noise variance and average error probability over an AWGN channel.

9. REFERENCES

1. Haykin, S. (2001). *Communication Systems* (4th ed.). John Wiley & Sons.
2. Proakis, J. G., & Salehi, M. (2008). *Digital Communications* (5th ed.). McGraw-Hill.
3. Lathi, B. P., & Ding, Z. (2009). *Modern Digital and Analog Communication Systems* (4th ed.). Oxford University Press.
4. Lab Manual: Digital Communication Laboratory (EC 39002), Spring 2025, IIT Kharagpur. (Base Reference - Implicit)

APPENDIX A: IC PIN DIAGRAMS

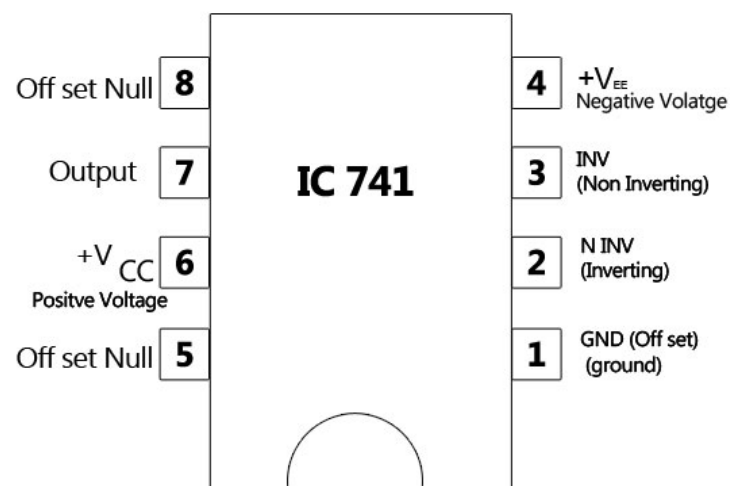


Fig A.01: Pin Diagram of IC741

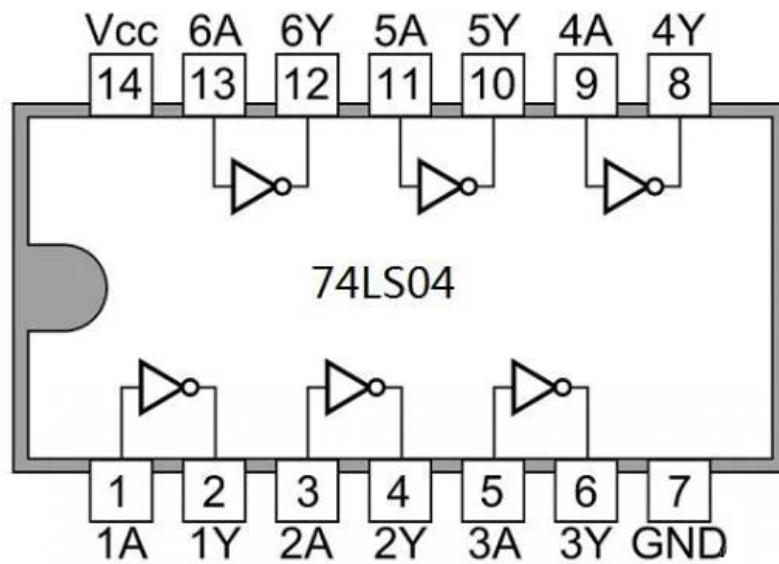


Fig A.02 : Pin Diagram of IC7404

7486 Quad 2-input ExOR Gates

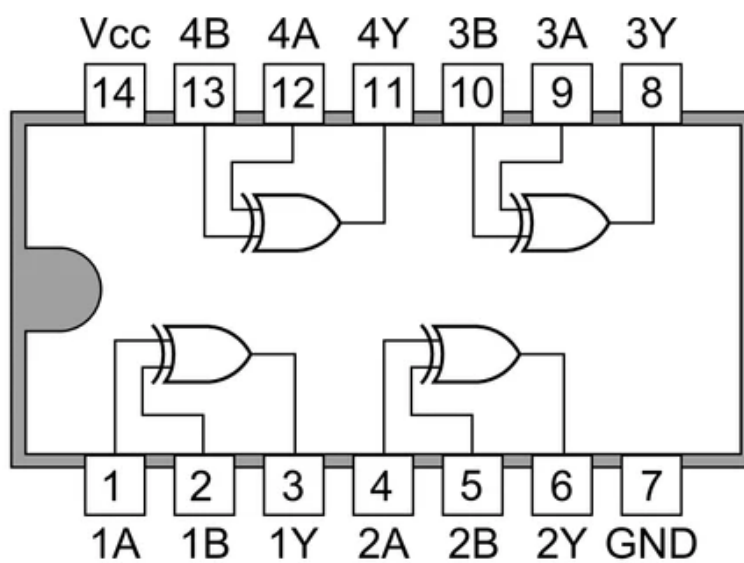
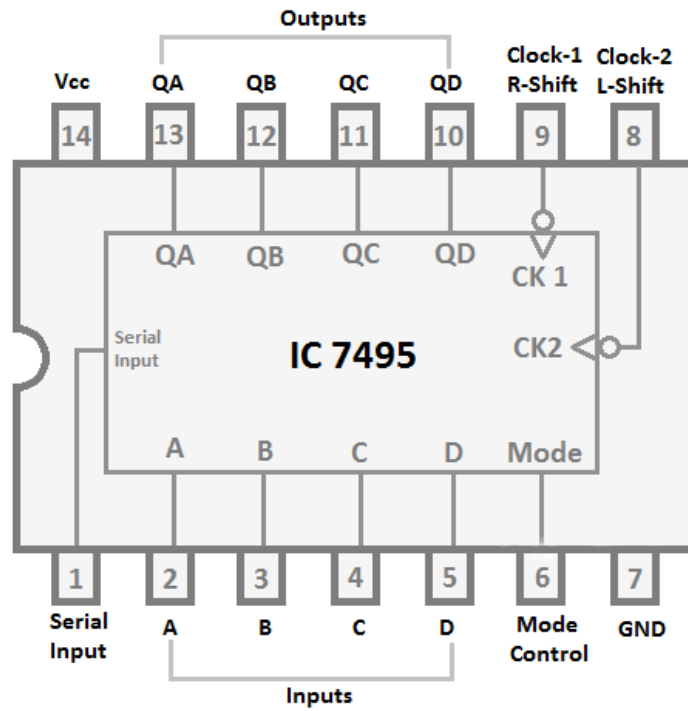


Fig A.03 : Pin Diagram of IC7486



IC 7495 Pin Diagram

Fig A.04 : Pin Diagram of IC 7495

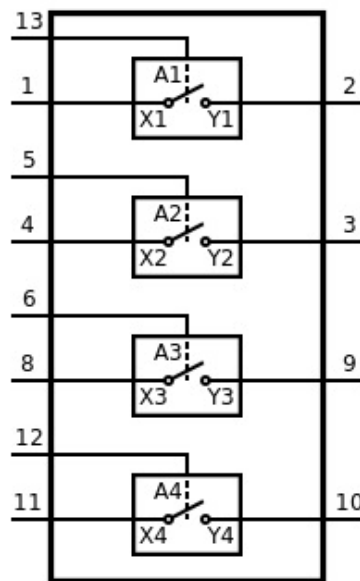


Fig A.05 : Pin Diagram of ICM 4016 (Switch)

APPENDIX B: MATLAB SIMULATION CODE

```
clc; clear; close all;
```

```

opts = detectImportOptions('bpsk_mod.csv');

opts.DataLines = [15 Inf];

data = readmatrix('bpsk_mod.csv', opts);

t = data(:, 1);

bpsk_sig = data(:, 2);

orig_msg = data(:, 3);

fs = 1 / (t(2) - t(1));

bpsk_sig = bpsk_sig - mean(bpsk_sig);

fc = 4000;

carrier_ref = sin(2*pi*fc*t);

demod_mixed = bpsk_sig .* carrier_ref;

window_size = round(fs / fc);

demod_lpf = movmean(demod_mixed, window_size);

recovered_data = demod_lpf > 0;

figure;

subplot(3,1,1);

plot(t, orig_msg);

title('Original Data');

subplot(3,1,2);

plot(t, bpsk_sig);

title('BPSK Signal');

subplot(3,1,3);

```

```

plot(t, recovered_data);

title('Recovered Data');

variances = linspace(0.01, 2.0, 15);

N = 256;

ber = zeros(length(variances), 1);

for i = 1:length(variances)
    var = variances(i);

    err = 0;

    for k = 1:N
        noisy_bpsk = bpsk_sig + sqrt(var) * randn(size(bpsk_sig));
        noisy_mixed = noisy_bpsk .* carrier_ref;
        noisy_lpf = movmean(noisy_mixed, window_size);
        noisy_recovered = noisy_lpf > 0;

        err = err + sum((orig_msg(100:end-100) > 2.5) ~= noisy_recovered(100:end-100));
    end

    ber(i) = err / (N * length(orig_msg));
end

figure;

semilogy(variances, ber, '-o');

title('BER vs Noise Variance');

xlabel('Noise Variance');

ylabel('BER');

grid on;

```